The S_3-symmetric 3HDM scalar sector: two corrections to our setup, and an insufficiency in the boundedness conditions we import from the literature

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Abstract

I have verified the S_3-symmetric three-Higgs-doublet scalar sector end to end, rebuilding the potential, the vacuum, the mass matrices and the parameter scan from the Lagrangian rather than from transcribed formulae. Two of our own inputs needed correcting. Separately — and this is the substantive result — the boundedness-from-below conditions we import from Das & Dey [1] are *necessary but not sufficient*: their Eq. (4g) is exactly the statement $f(1) > 0$ for a quartic $f(t)$ that must be non-negative for *all* $t \geq 0$. I give an explicit counterexample, derive the corrected condition in closed form, and find that it removes 9.5% of the parameter points Eq. (4) admits. The one caveat that governed how much of this was new is now settled: the [1] Erratum corrects only the tadpole equations and leaves Eq. (4) untouched (§4.6), so this is an insufficiency of the conditions as they stand in the final published record.

1 Summary

Everything below comes from `research/thdm_s3/01_scalar_parameter_space.ipynb`, which builds the model with `feynlag` — declaring the fields and their S_3 representations, then letting the library check invariance, solve the tadpoles, diagonalise and extract the spectrum. Nothing is transcribed from a paper except where explicitly marked as imported, and every step carries an independent check.

1. **Our aligned vacuum sat at $v = 155.03$ GeV, not 246.** Imposing the $\sqrt{3}$ alignment as a substitution silently overwrote v_1 and broke the electroweak-scale constraint. Every mass we quoted from that benchmark is low by a factor 1.587 (§3.1).
2. **Our λ basis is not [1]'s, and the difference is one sign.** The dictionary is now *solved* rather than assumed, from an over-determined system matching our independently derived masses to the closed forms of [2]. It comes out $e = -\lambda_4$, and the exercise doubles as an end-to-end validation of our whole chain against published results (§3.2).
3. **[1] Eq. (4) is not sufficient for boundedness** (§4) — the main result.
4. **Boundedness binds; unitarity does not.** Over $\lambda \in U(-2, 2)^8$, tree unitarity removes *no* points that boundedness has not already removed (§5).
5. **The 125 GeV condition is now a coupling cut, not a mass coincidence.** The state carrying the hVV coupling must sit at 125 ± 3 GeV, and light, appreciably coupled neutral states are vetoed. This keeps 64 of the 322 points a mass-only cut would keep. Over 6×10^7 samples, 1964 points survive (§5, §6).

Revised to replace the earlier mass-only 125 GeV cut with the coupling cut, and to quote the full 6×10^7-point scan. The main result (§4) is unchanged. Companion reports cover the fermion sector, the scalar decays and LFV rates, and the soft-broken vacuum.

2 The model and our conventions

Three Higgs doublets with $(SU(2)_L, U(1)_Y) = (\mathbf{2}, \frac{1}{2})$; under S_3 the pair (H_1, H_2) is a doublet and H_S a singlet. Writing $x_{ij} \equiv H_i^\dagger H_j$ for $i, j \in \{1, 2, S\}$, the Clebsch–Gordan decomposition $\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{1}' \oplus \mathbf{2}$ gives the structures

$$\underbrace{x_{11} + x_{22}}_{\mathbf{1}}, \qquad \underbrace{x_{12} - x_{21}}_{\mathbf{1}'}, \qquad \underbrace{d_2 = \big(x_{11} - x_{22},\, -(x_{12} + x_{21})\big)}_{\mathbf{2}}, \qquad x_S \equiv (x_{S1}, x_{S2}),$$

and the most general renormalisable S_3- and gauge-invariant potential

$$\begin{aligned} V = {} & \mu_1^2\,(x_{11} + x_{22}) + \mu_0^2\, x_{SS} + \lambda_1 (x_{11} + x_{22})^2 + \lambda_2 (x_{12} - x_{21})^2 + \lambda_3\, |d_2|^2 \\ & + \lambda_4 \big[x_S \cdot d_2 + \text{h.c.}\big] + \lambda_5\, x_{SS}(x_{11} + x_{22}) + \lambda_6 \big[x_{S1} x_{1S} + x_{S2} x_{2S}\big] + \lambda_7 \big[x_{S1}^2 + x_{S2}^2 + \text{h.c.}\big] + \lambda_8\, x_{SS}^2 . \end{aligned} \tag{1}$$

Three parametrisations of this same potential are in play and must not be conflated: ours $(\lambda_1 \ldots \lambda_8)$, [1] Eq. (3c) (also called $\lambda_1 \ldots \lambda_8$, but *not* the same objects), and [2] Eq. (2)/(6) $(a, \ldots, h)$. §3.2 settles the map. This matters because the constraints we want to apply are stated in [1]'s basis.

3 Two corrections to our own setup

3.1 The aligned vacuum sits at the wrong scale

The three tadpole conditions $\partial V / \partial v_i = 0$ are three equations for two free mass parameters (μ_0^2, μ_1^2). The system is over-constrained, and its residual vanishes only on $v_2 = \sqrt{3}\, v_1$ — so the alignment is a *consequence* of the model, not an input. That much is right.

What went wrong is arithmetic. Our benchmark picked $(v_1, v_2, v_S) = (200, 115, 80)$ GeV so that $\sqrt{\sum v_i^2} \approx 246$ GeV, then imposed the alignment as the substitution $v_1 \to v_2/\sqrt{3}$, which *replaces* $v_1 = 200$ by $115/\sqrt{3} = 66.4$ and never rechecks the scale. The aligned vacuum is $v = 155.03$ GeV. Since every mass^2 is homogeneous of degree 2 in the VEVs at fixed λ and θ, every mass we quoted is low by

$$246/155.03 = 1.587.$$

The fix is to impose both conditions simultaneously, which leaves exactly one free parameter — the angle θ of [2] Eq. (24) — with $v_{12} = v \sin\theta$, $v_S = v \cos\theta$, and the alignment then fixing $v_1 = v_{12}/2$, $v_2 = \sqrt{3}\, v_{12}/2$. This is now enforced in code and pinned by a regression test.

The qualitative conclusion drawn from that benchmark ("5 of 7 states tachyonic") is unaffected — but see §4: that point was never an admissible theory in the first place, because its potential is unbounded below.

3.2 The dictionary to [2], solved rather than guessed

Under a general $SO(2)$ rotation of the S_3 doublet (H_1, H_2), the contraction x_S rotates by α while d_2 rotates by 2α. Testing all eight quartic structures symbolically, **seven are invariant** — they

are the same object in either basis, so their coefficients transfer by term matching — and only $x_S \cdot d_2$, the λ_4 structure, can change. The only S_3-preserving rotation that changes it is $\alpha = \pi$, i.e. $H_{1,2} \to -H_{1,2}$, which flips its sign and leaves the other seven alone. So the entire dictionary is fixed up to one sign.

That sign is then determined, not chosen: [2] Eqs. (30)–(33) publish closed forms for the two pseudoscalar and two charged masses, we derive the same four independently, and equating them at several vacuum angles gives twenty linear equations for five unknowns. The over-determined system has a unique solution:

$$a = 2\lambda_8, \quad b = \lambda_5, \quad c = 2\lambda_1, \quad d = 2\lambda_2, \quad \boxed{e = -\lambda_4}, \quad f = \lambda_6, \quad g = 2\lambda_3, \quad h = 2\lambda_7.$$

Choosing $e = +\lambda_4$ instead fails *all four* published masses, so the sign is not marginal. That an over-determined system has any solution at all is the real content here: it validates our potential, tadpoles, mass matrices and rotation simultaneously against independently published closed forms.

Because the sign flip is a field redefinition, no physical condition may depend on it — and indeed λ_4 enters the constraints only as $|\lambda_4|$ or λ_4^2, so they transfer unchanged.

4 Main result: [1] Eq. (4) is not sufficient

4.1 The real neutral slice reduces to a quartic in one variable

Boundedness from below is directly testable: V_4 is homogeneous of degree 4, so the potential is bounded iff $V_4 \geq 0$ on the unit sphere in field space. Restricting to the real neutral directions $(H_1^0, H_2^0, H_S^0) = (r_1, r_2, r_S)$ and grouping,

$$V_4 = (\lambda_1 + \lambda_3)\, x^2 + (\lambda_5 + \lambda_6 + 2\lambda_7)\, xy + \lambda_8\, y^2 + 2\lambda_4\, r_S\, r_1 (r_1^2 - 3r_2^2), \qquad x = r_1^2 + r_2^2, \;\; y = r_S^2. \tag{2}$$

Two features of (2) are worth stating. First, λ_2 has *dropped out*: it multiplies $(x_{12} - x_{21})^2$, which needs a relative phase between the two doublets, and a real slice cannot see it. Second, the last term is the S_3 cubic covariant — in polar form $r_1 = \sqrt{x}\cos\psi$, $r_2 = \sqrt{x}\sin\psi$ it is exactly $2\lambda_4 \sqrt{y}\, x^{3/2} \cos 3\psi$, so its worst case over ψ is $-2|\lambda_4|\, x^{3/2}\sqrt{y}$. That half-integer power is what breaks any attempt to treat V_4 as a quadratic form in (x, y).

Every term is degree 2 in (x, y), so setting $x = 1$ and $t = \sqrt{y}$ loses nothing, and the **exact** condition on this slice is

$$\boxed{f(t) = \lambda_8 t^4 + (\lambda_5 + \lambda_6 + 2\lambda_7)\, t^2 - 2|\lambda_4|\, t + (\lambda_1 + \lambda_3) \;\geq\; 0 \qquad \text{for all } t \geq 0} \tag{3}$$

4.2 Eq. (4g) is precisely $f(1) > 0$

[1] Eq. (4g) reads

$$\lambda_1 + \lambda_3 + \lambda_5 + \lambda_6 + 2\lambda_7 + \lambda_8 > 2|\lambda_4|,$$

which is (3) evaluated at $t = 1$ and nowhere else. It tests one point of a curve that must stay non-negative everywhere. Necessary, not sufficient — and Fig. 1 shows what that costs.

4.3 An explicit counterexample

At

$$\lambda = (1.8057,\, -1.8434,\, -1.5837,\, -1.5192,\, 0.9116,\, 1.0769,\, -0.1231,\, 1.4658) \tag{4}$$

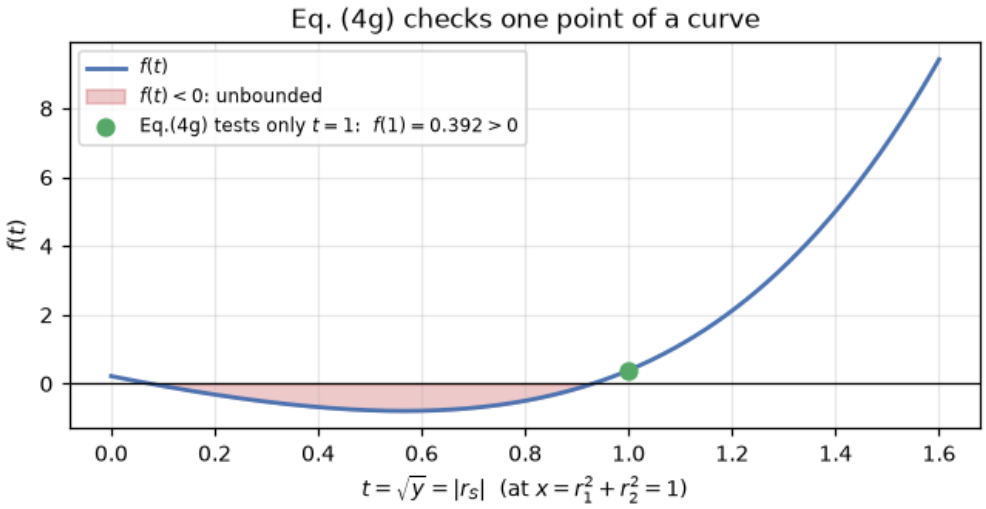


Figure 1: The quartic (3) at the counterexample point (4). Eq. (4g) tests only $t = 1$, where $f(1) = 0.392 > 0$ and the condition passes; the curve dips to -0.789 at $t = 0.565$, so the potential is unbounded below along a real neutral direction.

all seven conditions Eq. (4a)–(4g) are satisfied, yet along the unit direction

$$(r_1, r_2, r_S) = (-0.4598,\ -0.7944,\ 0.3970)$$

one finds $V_4 = -0.5074$, so $V_4 \to -\infty$ as the field norm grows. The closed-form minimum of (3) agrees with a direct scan to four decimals: both give -0.7891 at $t = 0.565$. This is checkable by hand in a few minutes, which is why it is quoted in full.

4.4 The corrected condition, and how much it removes

Minimising (3) over $t \geq 0$ in closed form (via the companion matrix of f', batched over parameter points) gives a condition that is exact on this slice. Its impact:

sample	Eq. (4) survivors	removed by the corrected condition
3×10^5 points	17 277	9.35%
2×10^6 points	114 293	9.47%

Two independent samples, consistent. Validation: over 400 probe points the closed form and a brute-force scan over 60 000 random directions give *zero* disagreements about the sign; at 40 000 directions both flag exactly the same 30 points. The closed form is not finding artefacts, and the sampling is not missing violations at this resolution.

4.5 Scope, stated honestly

Condition (3) is exact on the *real neutral* slice only. Complex-neutral and charged directions are not covered analytically; the notebook handles them with a numerical backstop over 40 000 directions in the full 12-dimensional field space. Over the full 6×10^7-point scan it removes 23 of the 1987 points that pass every other cut (none of the 64 in the two-million-point sample of §5). The residual gap is therefore real but small once the other cuts are applied. Doing this properly — separate analytic BFB-n and BFB-c conditions — is what [6] does for $U(1) \times U(1)$, $U(1) \times Z_2$ and $Z_2 \times Z_2$; S_3 is not among their cases, and that is the natural next piece of work.

4.6 The erratum, and why it does not affect this

[1] carries an **Erratum, Phys. Rev. D 91, 039905 (2015)**, never posted to arXiv — the arXiv record stops at v2 — which is why it went unread for so long. It is freely readable at APS, and it has now been read. Its entire content is a correction to **Eqs. (9a)–(9c)**, the tadpole (minimisation) conditions, closing with

> "These changes, however, do not affect any of our conclusions in the original paper."

Eq. (4) is untouched, and so are the unitarity eigenvalues Eqs. (36)–(37). Two consequences follow. First, the insufficiency demonstrated above is an insufficiency of the conditions in the *final published record* — not of a superseded pre-erratum form, and not a rediscovery of a correction already made. Second, our transcription of Eq. (4) and Eq. (37) is of the definitive text.

There is a pleasing footnote. The one equation Das & Dey had to correct is the one equation we never imported: our tadpoles are derived from the Lagrangian (§3.1), so the error could not reach us. Better, the *corrected* Eq. (9) is now an independent check: setting (9a) = (9b), since both equal $2\mu_1^2$, leaves $3\lambda_4 v_3(v_1^2 - 3v_2^2)/v_2$ and forces $v_1 = \sqrt{3}\,v_2$ — our own alignment, with $v_1 \leftrightarrow v_2$ exactly as the documented basis difference requires.

5 The scan

Two million λ points sampled uniformly on $U(-2,2)^8$ (seed `20260729`), with θ scanned at $v = 246$ GeV. The last three rows are the collider-facing cuts of §6:

cut	surviving	of previous
sampled	2 000 000	—
[1] Eq. (4) bounded	114 293	5.71%
+ corrected BFB (3)	103 465	90.5%
+ tree unitarity	103 465	100.0%
+ no tachyons	7 608	7.35%
+ $m_{H^\pm} > 80$ GeV [4]	6 890	90.6%
+ hVV carrier at 125 ± 3 GeV, $\kappa^2 \geq 0.9$	64	0.93%
+ LEP neutral veto [5]	64	100.0%
+ 12-dimensional BFB backstop	64	100.0%

A mass-only "one CP-even state at 125 ± 3 GeV" cut, which this report originally used, would keep 322 points at the sixth row. The sample is chunk 0 of the cached scan (`scan.py`). Swept over 30 such samples (6×10^7 points) the same cuts leave **1964** points (`results/viable_points.json`), and chunk 0 reproduces this table point for point.

The line worth noticing is tree unitarity: it removes **nothing**. In this λ range boundedness is the binding constraint. That is not because the condition is toothless — it rejects 5% of points at $|\lambda| \leq 4$ and half of them at $|\lambda| \leq 6$ — but because $|\lambda| \leq 2$ sits just inside it: the largest eigenvalue anywhere in the scan is 32.5, or 65% of the $16\pi \approx 50.3$ bound. (The much-quoted 3.63 belongs to the old benchmark $\lambda_k = 0.05k$, a single point, not to this sample.) The margin is real but narrower than that figure suggests, and it is worth knowing before spending effort on the unitarity side.

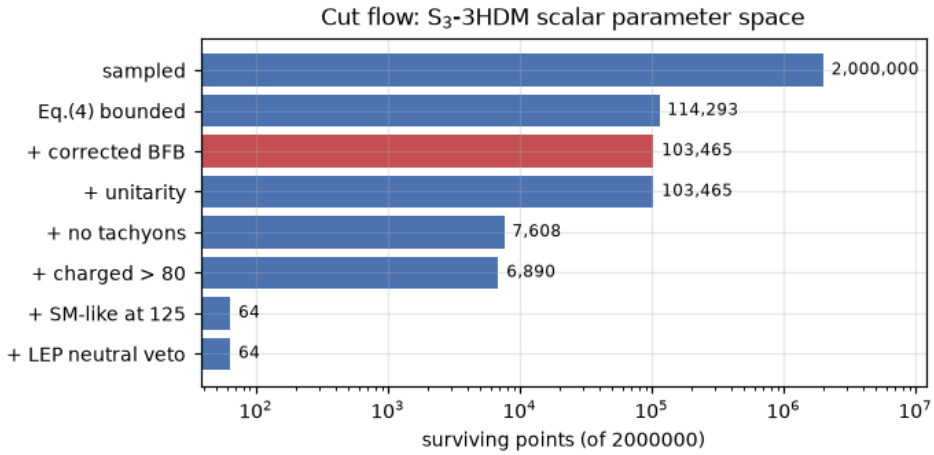


Figure 2: Cut flow over the 2×10^6-point scan, up to the theory cuts and a mass-only 125 GeV condition. The coupling-based rows that replace the last bar are in the table above.

6 What the viable region looks like

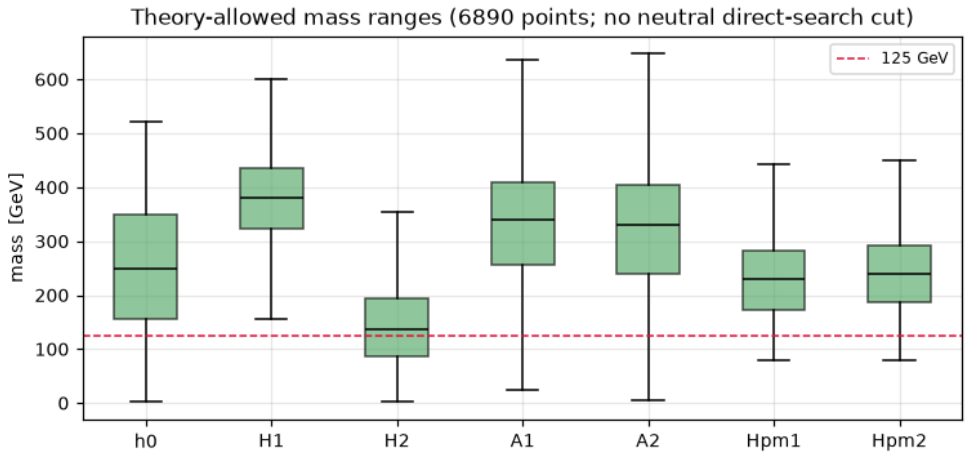


Figure 3: Theory-allowed mass ranges for the seven physical scalars over the 6890 points that pass boundedness, unitarity, no tachyons and $m_{H^\pm} > 80$ GeV, *before* the 125 GeV and neutral-state cuts below.

The theory-allowed spectra cluster below a TeV, consistent with [1]'s own conclusion that many new scalars must be light.

The collider cuts. The scan originally applied a mass-only condition (one CP-even state at 125 ± 3 GeV) and no bound at all on the neutral states. Both gaps are now closed with the hVV couplings, which follow from the same CP-even block as the masses:

- **SM-like at 125 GeV**: the state carrying the largest hVV coupling must sit at 125 ± 3 GeV with $\kappa^2 \geq 0.9$ of the SM strength. Over the points that pass the mass condition, only a minority put the coupling on the 125 GeV state, so the mass-only cut was too loose by a factor of five (322 against 64).
- **LEP neutral veto**: no CP-even state below 114.4 GeV may have $\xi^2 \geq 0.05$, a step approximation of [5] Fig. 10. The approximation carries no weight. h_0 is gauge-phobic exactly, for

any mixing, and the other two states share the SM strength, so $\kappa^2 \geq 0.9$ already forces the non-SM one below 0.1. The surviving set is the same for any floor in $[0.01, 0.10]$.

A light h_0 therefore passes on the physics ($g_{h_0VV} = 0$), not on an exemption: 117 of the 1964 points have $m_{h_0} < 114.4$ GeV.

Five representative points from the 1964, spanning the vacuum angle:

θ	m_{h_0}	m_{H_1}	m_{H_2}	m_{A_1}	m_{A_2}	$m_{H_1^\pm}$	$m_{H_2^\pm}$	SM-like	κ^2
0.438	318	127	109	253	188	202	127	H_1	0.993
0.712	281	328	125	305	326	256	267	H_2	0.999
0.859	32	261	127	339	272	197	211	H_2	0.922
1.040	345	245	127	484	370	310	194	H_2	0.995
1.459	102	322	127	472	383	231	345	H_2	0.940

All masses in GeV, at the 2nd, 25th, 50th, 75th and 98th percentiles of θ. κ^2 is the SM-like state's hVV coupling^2 in SM units. In the first row the non-SM H_2 at 109 GeV has $\xi^2 = 0.007$, under the LEP floor. In the third, the 32 GeV h_0 is gauge-phobic.

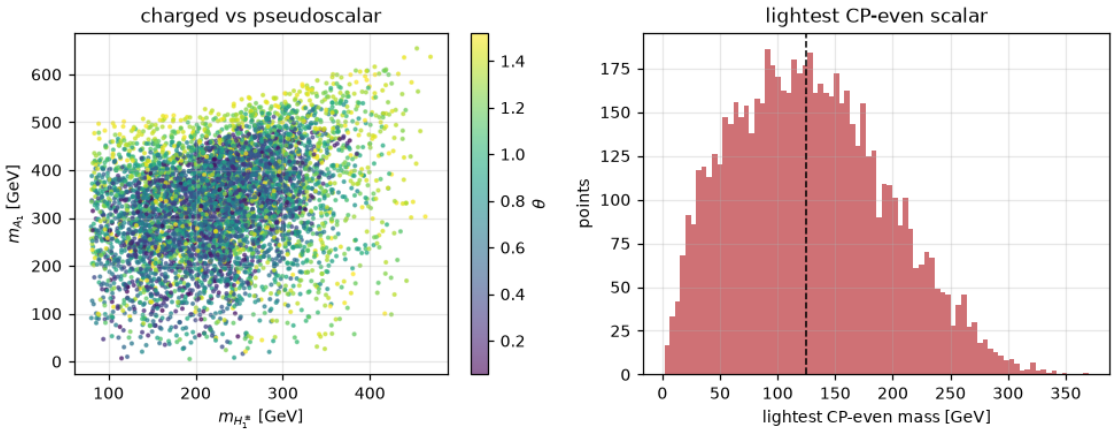


Figure 4: Correlations across the theory-allowed region (the 6890 points of Fig. 3).

7 Open questions, and what would help

1. **The neutral direct-search gap is closed.** The scan now cuts on the hVV coupling (§6). The couplings themselves, the decays and the LFV rates are in the companion decays report.
2. **The erratum question is closed.** Phys. Rev. D **91**, 039905 (2015) corrects Eqs. (9a)–(9c) only, and says so itself (§4.6). §4 is therefore not a rediscovery.
3. **Cross-check the unitarity bounds against [3]**, which recomputes them for all symmetry-constrained 3HDMs. Eq. (37) is now known to be final, but we still only check it for *internal* consistency; an independent recomputation would close that gap properly.
4. **Complete the boundedness analysis** for complex-neutral and charged directions, following [6]'s method. S_3 is not among the cases they treat.
5. **Is §4 worth writing up separately**, or does it belong as a section of the LFV paper? This is now the open question that matters. The constraint is widely used, the erratum does not fix it, and the corrected condition is a short closed form — so my instinct is that it stands alone. Your call.

A The derivation ledger

Each derived step in the notebook records what was claimed, whether it was derived here or imported, which independent checks it survived, and what would falsify it. The full typeset version is the companion document `derivations_01.tex`; in summary:

claim	status	independent checks
The $(a \dots h) \leftrightarrow \lambda$ dictionary, $e = -\lambda_4$	derived	solved from an over-determined system, not guessed; the four solved entries agree with term matching; $e = +\lambda_4$ fails all four published masses; the sign question vanishes as $\lambda_4 \to 0$
The real neutral slice, and the absence of λ_2	derived	degree-4 homogeneous; the cubic covariant is $\mathbb{Z}_3$-invariant (the check was falsified on a broken sign, so it has teeth); $\lambda_4 \to 0$ reduces to a quadratic form; an independent $r_2 = 0$ hand slice reproduces it
[1] Eq. (4a)–(4g)	**imported**	tested directly against our own V_4: counterexamples found
The exact boundedness condition (3)	derived	agrees with 60 000-direction brute force on every probe point; Eq. (4g) is exactly $f(1) > 0$

References

[1] D. Das and U. K. Dey, *Analysis of an extended scalar sector with S_3 symmetry*, Phys. Rev. D **89**, 095025 (2014), arXiv:1404.2491, doi:10.1103/PhysRevD.89.095025. Erratum: Phys. Rev. D **91**, 039905 (2015) — not posted to arXiv, but freely readable at APS. It corrects Eqs. (9a)–(9c), the tadpole conditions, and nothing else; Eq. (4) and Eqs. (36)–(37) are unaffected.

[2] M. Gómez-Bock, M. Mondragón and A. Pérez-Martínez, *Scalar and gauge sectors in the 3-Higgs Doublet Model under the S_3-symmetry*, Eur. Phys. J. C **81**, 942 (2021), arXiv:2102.02800, doi:10.1140/epjc/s10052-021-09731-3.

[3] M. P. Bento, J. C. Romão and J. P. Silva, *Unitarity bounds for all symmetry-constrained 3HDMs*, JHEP **08** (2022) 273, arXiv:2204.13130, doi:10.1007/JHEP08(2022)273.

[4] ALEPH, DELPHI, L3, OPAL Collaborations and the LEP Working Group for Higgs Boson Searches, *Search for Charged Higgs bosons: Combined Results Using LEP Data*, Eur. Phys. J. C **73**, 2463 (2013), arXiv:1301.6065, doi:10.1140/epjc/s10052-013-2463-1. The 80 GeV bound is the Type-II 2HDM value, applied here as a benchmark.

[5] ALEPH, DELPHI, L3, OPAL Collaborations and the LEP Working Group for Higgs Boson Searches, *Search for the Standard Model Higgs boson at LEP*, Phys. Lett. B **565**, 61 (2003), hep-ex/0306033, doi:10.1016/S0370-2693(03)00614-2.

[6] R. Boto, J. C. Romão and J. P. Silva, *Bounded from below conditions on a class of symmetry constrained 3HDM*, Phys. Rev. D **106**, 115010 (2022), arXiv:2208.01068, doi:10.1103/PhysRevD.106.115010.